[INCH-POUND]
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24 January 2000
SUPERSEDING
MIL-W-80037C
7 August 1980

COMMERCIAL ITEM DESCRIPTION

WELDING MACHINES, ARC, DC, GENERATOR, CONSTANT CURRENT, INTERNAL COMBUSTION ENGINE DRIVEN

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-W-80037C for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for DC generator arc welding machines driven by internal combustion engines to be used for portable arc welding applications.

2. CLASSIFICATION. The welding machines shall be classified according to the types and sizes specified in table I. The type and size of machine to be furnished shall be specified in the acquisition order (see 7.5(b)).

TABLE I. Arc welding machine classifications.

Drive Power	Cooling	Type	Size	Rating (Amperes) Minimum @ 100% duty cycle
Gasoline engine	Air	I	200	200
	Liquid	II	300	300
			350	350
Diesel engine	Liquid	III	300	300
			350	350
	Air	IV	250	250
			400	400
			600	600

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3431

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3. SALIENT CHARACTERISTICS

3.1 Components. The arc welding machine shall consist of an internal combustion engine, a DC electrical power generator capable of delivering constant direct current at levels suitable for arc welding, a frame, an enclosure, output terminals, and operating controls.

3.1.1 Unless otherwise specified (see 7.5(c)), the welding machine frame shall include a mounting structure which is arranged and pre-drilled for bolting the machine to a baseplate, skid, or trailer.

3.1.2 The machine frame shall include a lifting eye at the normal center of balance at the top of the machine. The lifting eye shall be capable of supporting two and one-half times the weight of the machine and shall be mounted so that either base-mounted or skid-mounted machines will retain alignment during lifting operations.

3.2 Performance. The output rating and performance capabilities of each machine shall comply with the Class I requirements of National Electrical Manufacturers Association (NEMA) Standard EW-1, "Electric Arc Welding Power Sources". Each machine shall be rated for a 100 percent duty cycle.

3.2.1 The welding machine open circuit voltage shall meet the requirements of NEMA EW-1.

3.2.2 The operating temperature limits of the welding machine generator and of all machine parts subject to damage by excessive temperature shall meet the requirements of NEMA EW-1.

3.3 Engine. The arc welding machine prime power source shall be a standard internal combustion engine of current commercial design and with sufficient horsepower to operate the welding power source at its maximum rated load and duty cycle.

3.3.1 Type I and II engines shall be 4-cycle gasoline engines designed to run on either leaded or unleaded gasoline with an octane rating of at least 83.

3.3.2 Type III and IV engines shall be either 2-cycle or 4-cycle diesel engines designed to run on No. 2 grade diesel fuel.

3.3.3 The arc welding machines shall contain a 12-volt electrical starting system with an automatic battery recharging capability. When specified, Type I machines may contain a manually operated starting mechanism rather than an electrical starting system.

3.3.4 Type II, III, and IV engines shall be equipped with a low oil pressure shutdown device.

3.3.5 The welding machine fuel storage capacity when the engine is operated at maximum load and at rated duty cycle shall be sufficient to operate Type I machines for at least three hours, Type II machines for at least six hours, and Type III and IV machines for at least eight hours.

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3.4 Output power source. The welding machine electrical power source shall meet the requirements of NEMA EW-1 and shall produce constant-current, self-regulating output power with a drooping volt-ampere characteristic.

3.4.1 Type II, III, and IV welding machines shall be provided with a switch to reverse the polarity of the welding current.

3.4.2 The welding current output shall be manually regulated in two steps. A coarse control shall select convenient output current increments through the entire range of the machine. A fine control shall then provide stepless adjustments within the range of each selected increment.

3.4.3 The machine shall contain an auxiliary power system meeting the requirements of NEMA EW-1. The auxiliary system shall include a 115-volt AC duplex receptacle circuit for lights and power tools. When specified, a 230-volt AC circuit with a designated kilowatt capacity shall also be provided (see 7.5(d)). Auxiliary circuits shall include automatic overload protection devices.

3.5 Cables. The welding cables and electrode holders shall be as specified in the contract (see 7.5(e)).

3.6 Nameplate. A nameplate conforming to the requirements of NEMA EW-1 shall be securely attached to the machine. Unless otherwise specified (7.5(f)), the nameplate shall display the following information:

a. Nomenclature
b. Manufacturer's type designation and/or identification number
c. NEMA EW-1
d. Maximum open circuit voltage
e. Rated load volts
f. Rated load amperes
g. Duty cycle at rated load
h. Maximum rated speed of generator in rpm at no load
i. Related frequency of power supply
j. Number of phases of power supply
k. Rated input voltage of power supply
l. National stock number (if specified)
m. Contract or order number (if specified)
n. Date of manufacture (if specified)

3.7 Trailer. When specified (see 7.5(g)), the welding machine shall be mounted on an over the road trailer. The trailer shall be capable of maintaining speeds of up to 50 mph on improved roads and 10 mph over rough terrain.

3.8 Accessories. All accessories and optional equipment shall be as specified in the contract (see 7.5(h)).

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The engine-driven arc welding machines provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and have been also offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

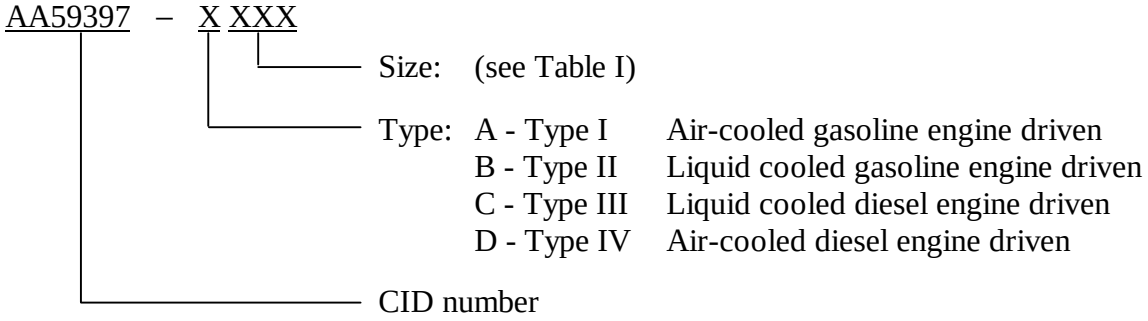
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the welding machine shall be preserved, packed, and marked as specified in the procurement contract or order (see 7.5(i)).

7. NOTES

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained form the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.

7.2.2 NEMA EW-1. Copies of NEMA Standard EW-1 may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

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7.3 National stock numbers (NSNs). The NSNs listed below are assigned for the arc welding machine types and sizes specified in this CID. Other NSNs may also apply to this document.

Type	Size	Assigned NSN
III	300	3431-00-708-7625
III	300	3431-00-846-9636

7.4 Sources of supply. The manufacturers and/or suppliers listed below are known to supply arc welding machines that meet the salient characteristics requirements of this CID. Competition is not limited to the listed firms.

L-TEC Welding and Cutting Systems
Florence, South Carolina 29501

Miller Electric Manufacturing Company
Appleton, Wisconsin 54914

Teledyne McKay
York, Pennsylvania 17405

Thermadyne Industries, Incorporated
Troy, Ohio 45373

7.5 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Classification (type and size) of arc welding machine (see table I)
c. Frame mounting structure, if different (see 3.1.1)
d. Auxiliary power requirements (see 3.4.3)
e. Welding cables (see 3.5)
f. Nameplate, if different (see 3.6)
g. Trailer, if required (see 3.7)
h. Accessories and optional equipment, if required (see 3.8)
i. Packaging requirements (see 6.1)

MILITARY INTERESTS:

Custodians:
Army - AV
Navy - YD
Air Force - 99

Reviewers:
Navy - MC, SH
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3431-0253)

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